

Supplementary file containing detailed experimental results for Reviewer 3rTN

Table 1. Neural architecture performance prediction results on the Nas-Bench-NLP dataset. Mean and standard deviation over 9 trials are reported. The best performance is underlined. ResGatedGCN is used as the backbone neural architecture encoder. Our method (FGP) achieves the highest scores across all evaluation metrics.

Methods\Metrics	Kendall's Tau	Recal@1%	Recall@5%
N/A (No pre-training)	28.5 (6.1)	2.9 (1.4)	14.8 (6.9)
GMAE	28.9 (6.6)	2.2 (1.0)	15.7 (5.3)
ZC-Proxy	28.7 (5.6)	2.9 (2.3)	15.5 (3.8)
FGP (Ours)	<u>29.7 (4.8)</u>	<u>4.3 (2.5)</u>	<u>16.2 (7.1)</u>

Table 2. Operation classification results. Mean and standard deviation over 10 trials are reported. In all the settings, an MLP trained with flow-surrogate-based features achieves more than 92% of accuracy.

Datasets \ Operations	1x1	3x3	Pooling
NAS-Bench-101	93.2 (0.3)	99.9 (0.2)	92.1 (0.7)
NAS-Bench-201	98.9 (0.3)	99.9 (0.1)	1.0 (0.0)

Table 3. Neural architecture performance prediction results under diverse pre-training data usage ratio. The mean and standard deviation of over 9 trials are reported. The best performance is underlined. ResGatedGCN is used as the backbone neural architecture encoder. Our method (FGP) achieves the highest scores across all evaluation metrics.

Metric: Kendall's Tau	Dataset: NAS-Bench-101			Dataset: NAS-Bench-201		
	GMAE	ZC-Proxy	FGP (Ours)	GMAE	ZC-Proxy	FGP (Ours)
20% of the training set is used for pre-training	67.6 (6.1)	65.1 (7.5)	<u>70.6 (4.6)</u>	75.2 (1.7)	73.8 (1.0)	<u>78.5 (0.8)</u>
40% of the training set is used for pre-training	67.9 (6.4)	66.0 (5.9)	<u>72.1 (3.9)</u>	76.2 (1.2)	74.3 (1.5)	<u>79.4 (1.2)</u>
60% of the training set is used for pre-training	68.8 (5.9)	66.3 (6.0)	<u>72.7 (4.4)</u>	76.9 (0.9)	74.8 (1.4)	<u>80.3 (0.9)</u>
80% of the training set is used for pre-training	69.1 (5.2)	67.4 (4.1)	<u>72.9 (5.1)</u>	77.7 (1.0)	74.5 (1.7)	<u>80.6 (0.8)</u>
100% of the training set is used for pre-training	69.8 (5.1)	67.3 (5.2)	<u>73.5 (5.5)</u>	79.8 (1.3)	75.2 (1.4)	<u>81.4 (0.7)</u>
100% of the training set and their random augmentations are used for pre-training	69.6 (5.2)	66.8 (5.4)	<u>73.0 (5.5)</u>	79.3 (1.5)	74.8 (1.5)	<u>80.6 (1.0)</u>

Table 4. Neural architecture performance prediction results using the new neural architecture encoder, DiGCN (graph neural network for directed graphs). Mean and standard deviation over 9 trials are reported. The best performance is underlined. Our method (FGP) achieves the highest scores across all evaluation metrics.

Methods\Metrics	Dataset: NAS-Bench-101			Dataset: NAS-Bench-201		
	Kendall's Tau	Recall@1%	Recall@5%	Kendall's Tau	Recall@1%	Recall@5%
N/A (No pre-training)	65.2 (4.4)	25.6 (5.5)	49.0 (1.1)	65.7 (1.5)	14.1 (4.8)	43.9 (2.6)
GMAE	65.1 (3.9)	26.3 (535)	49.5 (2.7)	66.0 (1.4)	16.2 (5.2)	44.2 (2.1)
ZC-Proxy	66.9 (4.3)	26.0 (3.9)	50.5 (5.7)	72.1 (1.3)	18.8 (4.7)	47.1 (1.9)
FGP (Ours)	<u>73.3 (3.7)</u>	<u>37.3 (8.6)</u>	<u>57.5 (5.6)</u>	<u>77.9 (2.0)</u>	<u>23.8 (8.1)</u>	<u>50.9 (5.8)</u>

Table 5. Neural architecture processing time (per architecture) and flow surrogate derivation time (per architecture) for different types of neural architectures. Results are similar for CNNs (NAS-Bench-101 and NAS-Bench-201) and for an LLM (Llama 3).

Computation time (seconds)	NAS-Bench-101	NAS-Bench-201	Llama 3 block
Neural architecture processing with ResGatedGCN neural architecture encoder	0.0021	0.0022	0.0025
Flow surrogate derivation	0.0022	0.0024	0.0026

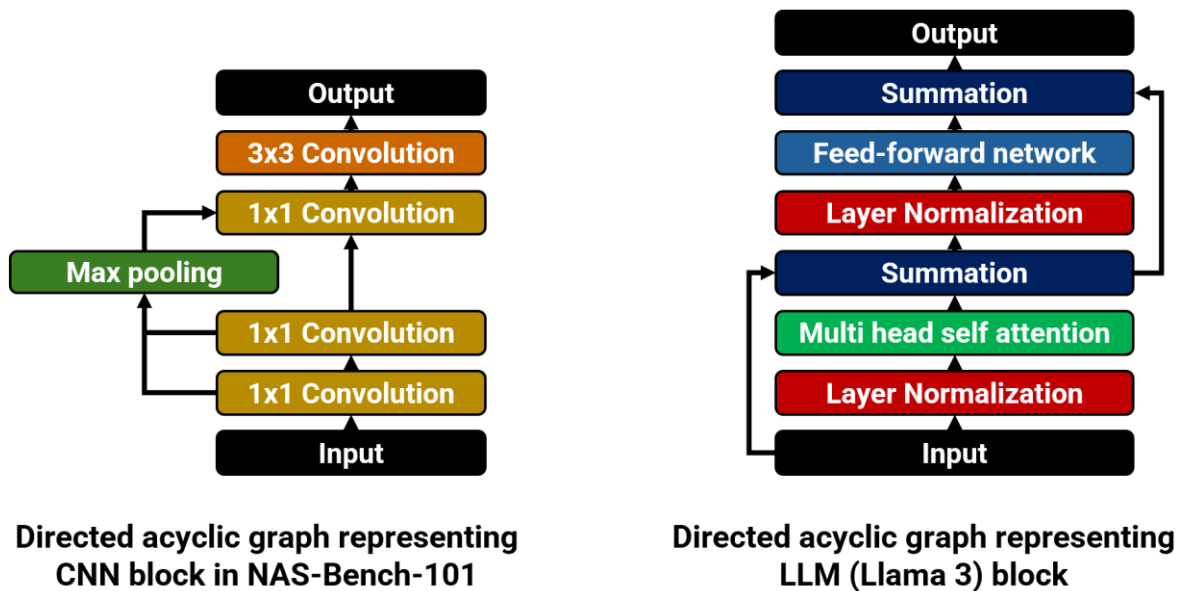


Figure 1. Visualization of directed acyclic graphs (DAGs) representing a CNN block (left) and an LLM block (right), respectively. Although the number of parameters differs significantly, the sizes of their operation-level DAGs are comparable.

Table 6. Neural architecture performance prediction results using different rounds of message passing. Mean and standard deviation over 9 trials are reported. The best performance is underlined. ResGatedGCN is used as the backbone neural architecture encoder. Our method (FGP) achieves the highest scores across all evaluation metrics.

Methods\Metrics	Dataset: NAS-Bench-101			Dataset: NAS-Bench-201		
	Kendall's Tau	Recall@1%	Recall@5%	Kendall's Tau	Recall@1%	Recall@5%
Use 1-round flow surrogates (FGP)	<u>74.8 (4.8)</u>	<u>37.5 (13.0)</u>	<u>61.7 (3.6)</u>	<u>82.2 (0.7)</u>	<u>48.9 (8.1)</u>	62.3 (2.5)
Use 1- and 2-round flow surrogates	71.7 (5.4)	36.7 (5.9)	58.8 (3.4)	80.6 (0.5)	46.0 (6.0)	62.0 (1.9)
Use 1-, 2-, and 3-round flow surrogates	70.7 (4.7)	34.0 (8.9)	58.2 (4.1)	81.6 (0.8)	47.3 (0.6)	<u>62.5 (1.0)</u>